

A 12,800-year-old layer with cometary dust, microspherules, and platinum anomaly recorded in multiple cores from Baffin Bay

PLOS One (2025) - 24 Comments

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#1 *Sinilabeo dero* comment accepted August 2025

The following statement is incorrect:

Some critics have claimed that YDB impact proxies were not formed during impact events but rather by other processes that coincidentally deposited them simultaneously [98,100,106,149].

with these references;

98: Boslough M, Nicoll K, Holliday V, Daulton T, Meltzer D, Pinter N. Arguments and evidence against a Younger Dryas impact event. In: Giosan L, Fuller D, Nicoll K, Flad R, Clift P, editors. Climates, landscapes, civilizations. Washington, DC: Am Geophys Union. 2012. p. 13–26. doi: 10.1029/2012GM001239.

100: Bourne AJ, Cook E, Abbott PM, Seierstad IK, Steffensen JP, Svensson A, et al. A tephra lattice for Greenland and a reconstruction of volcanic events spanning 25–45 ka b2k. Quaternary Sci Rev. 2015;118:122–41. doi: 10.1016/j.quascirev.2014.07.017.

106: Roberts DL, Matthews T, Herries AIR, Boulter C, Scott L, Dondo C, et al. Regional and global context of the Late Cenozoic Langebaanweg (LBW) palaeontological site: West Coast of South Africa. Earth-Sci Rev. 2011;106(3–4):191–214. doi: 10.1016/j.earscirev.2011.02.002.

149: Holliday VT, Daulton TL, Bartlein PJ, Boslough MB, Breslawski RP, Fisher AE, et al. Comprehensive refutation of the Younger Dryas Impact Hypothesis (YDIH). Earth-Science Reviews (2023) 247:105022. doi: 10.1016/j.earscirev.2023.105022

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Holliday et al (ref. 149) clearly do not state an opinion that remotely resembles that which the authors of this paper attribute to them. In fact, they explicitly reject the YDIH proposition that putative impact proxies are deposited simultaneously. Here are a few quotes from the paper that all contradict the quote from Moore et al.

To date, no section with thousands of years of continuous deposition has been subjected to close interval sampling to determine if suites of purported impact proxies are unique to the YDB.

No numerical age control for purported YDB with “impact markers.”

Firestone et al. (2006) misunderstand the difference between OSL and radiocarbon dates. They cite dates with low precisions of 10–50% and some dates are >12.k cal yr BP, while others are older.

Blackville, SC: OSL date of $12,960 \pm 1190$ yr BP is from spherule zone in sand rim along Bay on erosional disconformity with potential mixing of dated sediments. Dates have low OSL precision and OSL age reversal among the three published dates, but this is not discussed or addressed. Comment: No numerical age control for purported YDB with “impact”

Kimbel Bay, NC: Spherule zone 10 cm above the youngest OSL age of 25.5 ka in sand rim along Bay. No indication of depositional or weathering hiatus, nor erosion. The time gap not considered in age estimate nor discussed or otherwise considered. Age reversal not discussed. Comment: No numerical age control for purported YDB with “impact markers.”

the Pt anomaly appears in the ice core about 1 m above the YDB, which corresponds to about 20 years

There are many more similar point-by-point rejections by Holliday et al of coincident simultaneity. Their words explicitly contradict the statement that they argued that other processes “coincidentally deposited them [impact markers] simultaneously” and it is difficult to understand how anyone who has read their paper could think that this was their position.

Boslough et al (ref. 98, with many of the same authors) also clearly stated their rejection of coincident simultaneity of putative impact markers as follows:

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As stated by Firestone et al. [2007], the impact hypothesis is based on the interpretation of a purported carbon-rich "black mat" stratigraphic layer and its constituent minerals, which are presumed to represent a single horizon of the YD boundary. The term "black mat" refers to a wide array of late Quaternary wetland deposits and soils found in stream, lake, and pond settings, primarily across the Southwest and Great Plains of North America ranging in age from 6 to more than 40 ka [Pigati et al., 2012].

It is unclear how the citations 100 & 106 are relevant to the claim either. Perhaps all four citation numbers are typographical errors. I urge the authors to publish a quick correction to avoid the risk that these errors will be propagated into other published works by authors who do not check primary sources and could inadvertently misrepresent them without reading them.

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Reply

#2 *Sinilabeo dero* comment accepted August 2025

Supporters of the YDIH have adopted unconventional nomenclature, sometimes defining terms differently than the way they have been used in the mainstream literature on airbursts for decades (see for example, this comment (<https://pubpeer.com/publications/7F7A12BDAE1D37E5B0844E764CD0FF>)). These redefinitions confuse non-experts and also make it difficult for seasoned experts in airbursts, cratering impacts, and planetary defense to follow along.

The authors appeal to "touchdown" airbursts in several passages:

Based on this compositional data, we propose that Baffin Bay microspherules comprise terrestrial target materials melted by touchdown airbursts [114,115]

Our evidence suggests that YDB Baffin Bay microspherules, as with other YDB examples elsewhere, most likely formed from terrestrial material mixed with a small percentage of ET material that melted during low-altitude touchdown airbursts/impacts [114,115].

Van Ginneken [115] described evidence suggesting the occurrence of a type-2 or touchdown airburst in Antarctica, which involves the impactor's vaporized material reaching Earth's surface at hypervelocity [133].

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We thus conclude that Baffin Bay microspherules were produced from terrestrial material mixed with a few percent of the impactor that melted during near-surface airburst plumes (type-2 or touchdown airburst) [115] or crater-forming impacts that reach Earth's surface rather than as a result of meteoritic ablation or cosmic influx.

Such microtektites and microkrystites are known to have been deposited as proximal or distal ejecta from ET impacts and touchdown airbursts [13,14,114,115].

Some fragments exploded at high altitudes with little surface effects, while others exploded very close to the ground and are known as type-2 or touchdown airbursts [115,133,168,169].

Citations [13], [14], and [114] were published in 1990, 1987, and 2014, respectively. They are puzzling because none of these sources mentions touchdown airbursts, a term that wasn't coined until 2021 by Van Ginneken et al (<https://www.science.org/doi/10.1126/sciadv.abc1008>) (citation [115]). Citation [133] (Boslough et al, 2022 (<https://www.hou.usra.edu/meetings/crater2022/pdf/2021.pdf>)) is also puzzling, because it does not mention Antarctica. It does not use the term "touchdown airburst". It refers to such an event as a "Type 2 contact airburst" or "Fire on the Ground". It cites Boslough, (2014) (<https://www.osti.gov/servlets/purl/1106311>) for the "Type 1" and "Type 2" nomenclature. This abstract and most previous papers refer to the downward-directed hot meteoritic vapor as a "fireball" or "jet" and the material ejected upward as a "plume" ("fireball" was also used by some modelers of the Shoemaker-Levy 9 impacts for the incandescent stage of the plume, but "plume" has been used ever since by the mainstream airburst and cratering impact community of experts).

Citation [169] refers, ambiguously, to "Boslough M. In: 2015". Presumably this was intended to cite Boslough (2015) (<https://www.unm.edu/~mbeb/Publications/Boslough-airbursts-2015.pdf>). "Airburst Modeling." In: Handbook of Cosmic Hazards and Planetary Defense 665-692, which also makes no mention of "touchdown airbursts" (because that term would not be coined until 2021). But it does define Type 1 and Type 2 airbursts the same way as Boslough (2014). Below is the figure and definition from the 2015 "Airburst Modeling" chapter.



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Two types of Low-Altitude Airburst

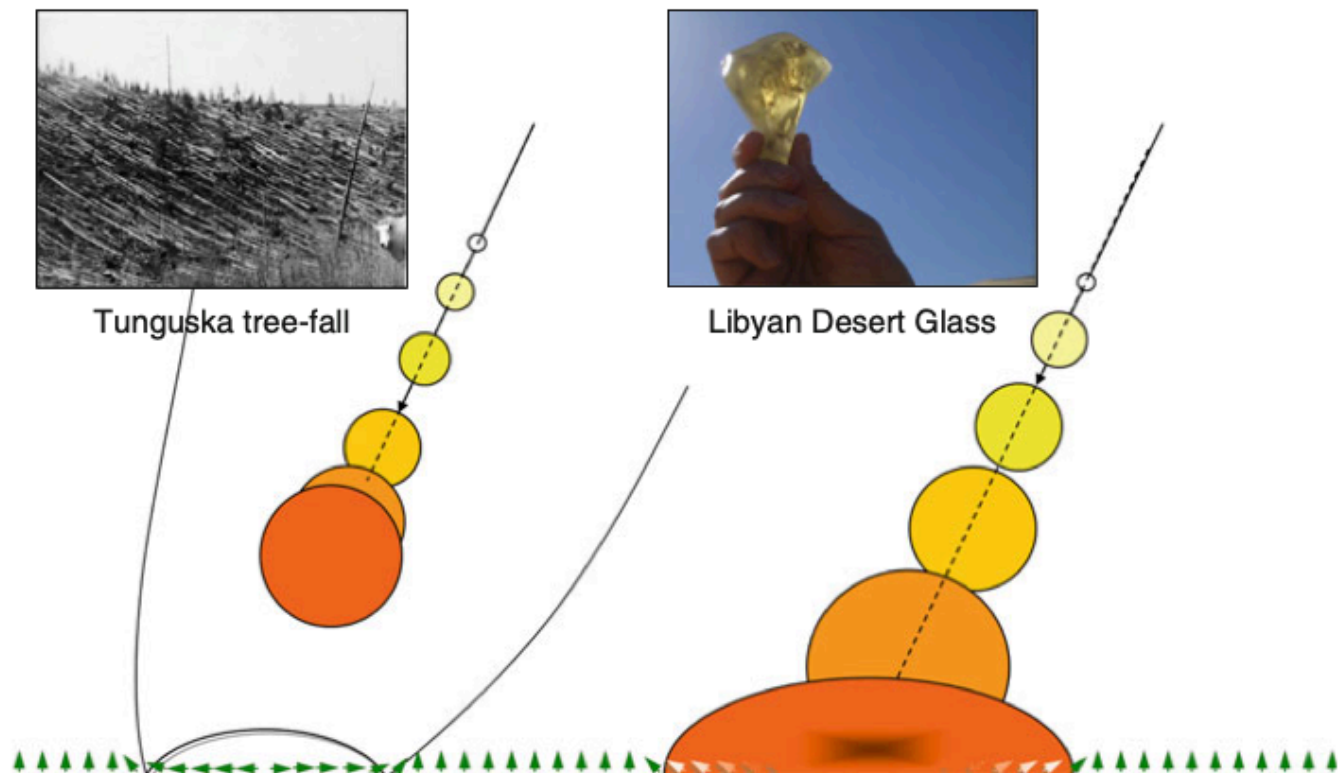


Fig. 3 Left: Type 1 airburst (Tunguska type). Right: Type 2 airburst (Libyan Desert Type)

cross-sectional area. At (c), the asteroid breaks into fragments which rapidly ablate, forming a high-temperature fireball. At (d), the fireball continues to drop to surface where it is held in contact for some period of time by its own inertia. This is consistent with observations of Libyan Desert Glass from an event in the western desert of Egypt 29 Ma ago. These have been defined as two distinct types of airbursts by Boslough (2014), as shown in Fig. 3 where a type 1 airburst (Tunguska type) is shown on the left, and a type 2 airburst (Libyan Desert type) is shown on the right. The following two sections describe the development of models for these two types of airbursts, drawing heavily on the original paper by Boslough and Crawford (2008).

(<https://pubpeer.com/storage/image-1754630595030.png>)

I recommend that the authors correct their citations. I assume they intended to cite [116] (van Ginneken M, Harvey RP, Goderis S, Artemieva N, Boslough M, and Boslough M. "The identification of airbursts in the past: Insights from the DDT-58 Google Earth web browser." *Earth and Planetary Science Letters* 2014; 297:162-171. doi:10.1016/j.epsl.2014.01.016).
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"touchdown airburst" to be another term for the previously defined Type 2 airburst". Since the citations need to be corrected, I urge the authors to correct their other terminology to be consistent with the words that mainstream researchers have been using for decades, to avoid creating further confusion.

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Reply

#3 *Actinopolyspora biskrensis* comment accepted August 2025

From the paper:

"Fe-rich microspherules exhibit blebs of a low-oxygen metal (e.g., Fig. 5d) that is transitional between chromite (**FeC₂O₄**; melting point = 1,850–2,200 °C) and Cr-magnetite (Fe²⁺(Fe_{1-n}Cr_n)₂O₄; melting point = 1,583–1,587 °C), both found in some chondritic and achondritic meteorites..." (page 6)

FeC₂O₄ is iron(II) oxalate, not chromite. Chromite's correct formula is FeCr₂O₄. Since I only found one occurrence for this, it probably is merely a typo (C₂ should be Cr₂). Could the authors please check?

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Reply

#4 Mark Boslough comment accepted August 2025

I am perplexed by the citations that are used in support of the statement, "This impactor is hypothesized to have undergone multiple disintegrations during the last 20,000–35,000 years, resulting in the formation of the Taurid complex, which includes Comet Encke and approximately 90 objects with diameters up to 5 km [2–7]" which appears the first section of the introduction, in which the Younger Dryas impact hypothesis (YDIH) is defined.

Two of the citations (3 & 5) present data from sky camera observations of Taurid meteors, which are phenomena associated with small objects that occur within Earth's atmosphere. These papers did not present any new observational data about any large objects in the Taurid Complex, which orbit the sun and require observations by large survey telescopes. Two more citations (4 & 6) are about observations of objects outside the solar system. Citation 4 is about Cygnus X-1, which is a black hole that is 7000 light years away and the authors do not explain how it could have any bearing on the Taurid complex. Citation 6 is about a high speed impact on the 1940s.

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hypothesis of Alexander Kazantsev. Citation 6 provides observational data of the Andromeda galaxy, which is more than 150,000 light years away and its connection to our solar system (or even to our galaxy) is also never stated.

Citations (2) & (7), on the other hand, do mention objects that might be associated with the Taurid complex, but neither of the papers support the statement that it includes “approximately 90 objects with diameters up to 5 km”. Citation (2) includes a table of NEOs with all orbital elements with the exception of longitude of the perihelion similar to the TC. The fifteen that have a perihelion longitude within a 90 degree band are listed as being “aligned with Taurids” and discussed in the text as being “apparently aligned” with a strong caveat:

it is important to acknowledge that the giant comet hypothesis is not the only explanation which might fit the observations, and the possibility that some form of selection effect has led to the apparent alignment of the objects that we call the Taurid Complex should certainly be entertained. Indeed, Valsecchi (1995, personal communication) has indicated that, along with coworkers, he has identified a discovery selection effect that might provide an alternative explanation for the apparent asteroid alignment in Table 1, the alignment with 2P/Encke and the Taurid meteoroids then being assumed to be a coincidence. Here's the figure from citation (1) that shows the set of orbits that have this *apparent* alignment based on their perihelion longitude.

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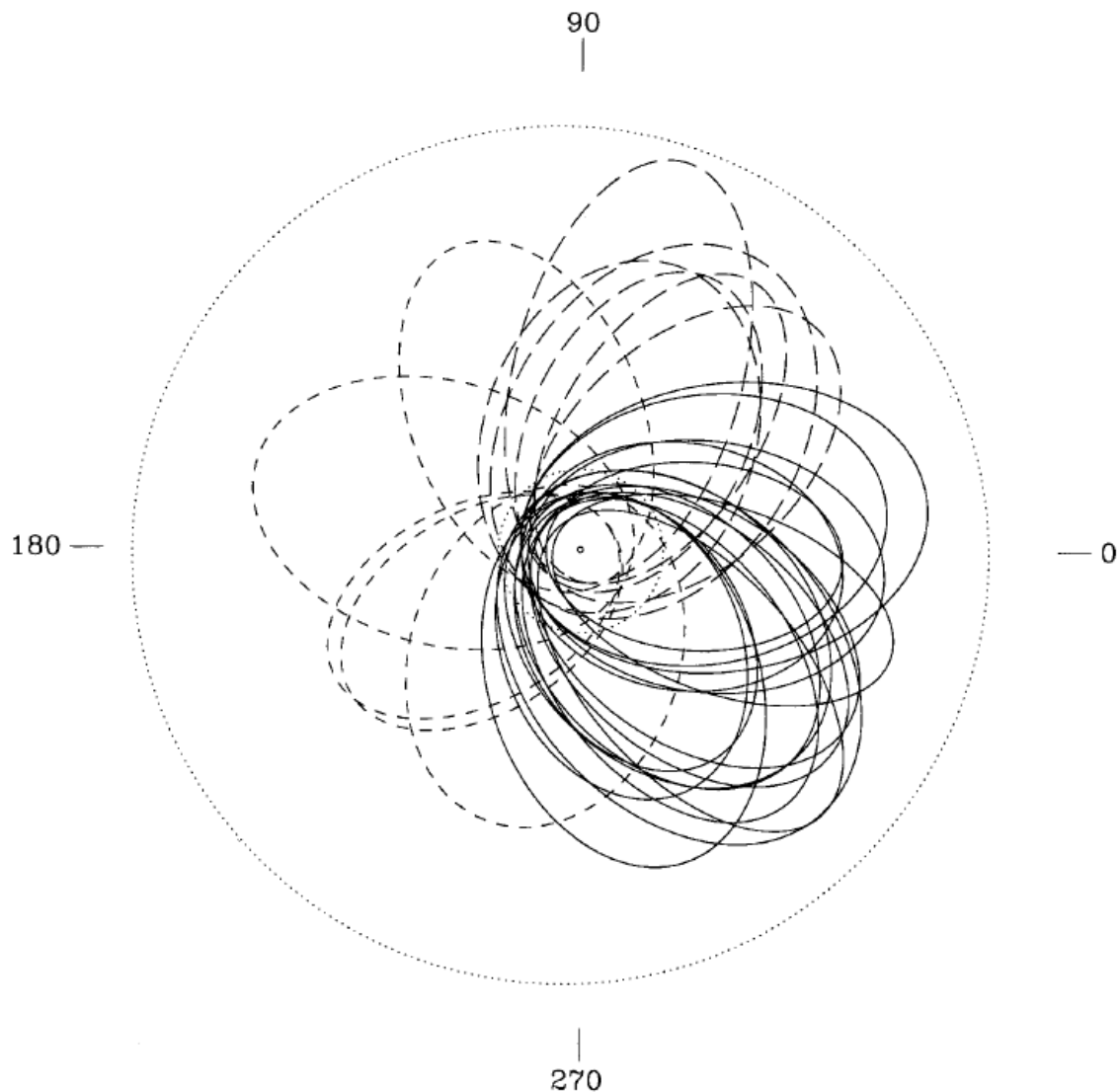


Figure 1. The orbits of all objects listed in Table 1, projected on to the ecliptic plane. The orbits of the Earth and Jupiter are shown as dotted lines. The 16 Taurid Complex objects are plotted as solid lines, the six Hephaisstos group objects as long-dashed lines, and the other five objects (all asteroids) as short-dashed lines. As indicated by the numbers, longitude increases anticlockwise (i.e., this is a view from north of the ecliptic), the origin being towards the right.

(<https://pubpeer.com/storage/image-1754857468986.png>)

The statement that the Taurid complex includes “approximately 90 objects with diameters up to 5 km” cannot be attributed to any of the first five citations on the list, and I suggest that a correction be published to remove them. It appears that the statement is solely attributable to a mistaken representation of citation (7), which shows a table of 88 (which is “approximately 90”) Taurid complex candidates (were the operative word is “candidates”), based on the authors’ modified D criterion that allows more candidates than the conventional D criterion. The arbitrariness of membership in a group such as the Taurid complex is explicitly stated by the authors of citation (7):

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It is worthwhile to note that the condition $D < D_c$ defines obviously a sharp boundary between members and non-members of a group, a situation that very rarely occurs in nature. Actually, mainly due the uncertainties on the orbital parameters of the objects and on the ones used as reference, the membership has to be assessed using a probabilistic approach. In other words, for a given object, the smaller is D compared to D_c , the greater is the probability that such object is a member of the TC; conversely, the larger D is than D_c , the greater the probability that that object is not a member. This being the case, it is not reasonable to exclude from the TC an object e.g. with $D = 0.27$. On the contrary, for an object e.g. with $D = 0.33$ one can say with an excellent degree of reliability that it is not a TC member. For these reasons, we consider all the objects with D between 0.25 and, let say 0.30 (an upper limit suggested by Porubcan et al., 2006), as probable members to the TC, with their membership to the complex that is more or less uncertain depending on the deviation of D from D_c .

(<https://pubpeer.com/storage/image-1754857667683.png>)

Moreover, none of the *candidates* identified as being as big as or bigger than 5 km in diameter. Therefore, the claim that the Taurid complex includes “approximately 90 objects with diameters up to 5 km” is not attributable to any of the six papers cited. I recommend that the journal editors ask the authors to make corrections to remedy this misunderstanding and improper citation, perhaps along with the corrections suggested in comments #1 & #2.

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Reply

#5 Mark Boslough comment accepted August 2025

A couple typos & points of confusion in my comment #4 have been pointed out to me by another PubPeer user.

The sentence “Here's the figure from citation (1) that shows the set of orbits that have this *apparent* alignment based on their perihelion longitude” was my own text, and was accidentally included in the quote block above

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The sentence “Moreover, none of the *candidates* identified as being as big as or bigger than 5 km in diameter” could have been stated more directly as “Citation (7) does not provide sizes for the 88 candidates, so the statement that they have “diameters up to 5 km” is unfounded.” © 2026 PubPeer Foundation (<http://www.pubpeer.com/>)

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Reply

#6 **Marc D. Young** comment accepted August 2025

Author Response

Due to issues with incorrect DOI numbers (PLoS One relies heavily on DOI regardless of what the written reference is), 43 references have been identified as incorrect, despite being correct on the submitted manuscript.

As such, a correction request will be issued to PLoS One which should remedy the comments here regarding apparently nonsensical citations. We will also fix the incorrect chemical formula identified by Elizabeth Bik above.

Thanks for your close attention and keen interest in our groundbreaking research!

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Reply

#7 **Marc D. Young** comment accepted August 2025

Author Response

#1

The anonymous poster Sinilabeo dero is incorrect, as they mistake the distinct concepts of synchronicity and simultaneity. Critics of our work argue against synchronicity of deposition between sites, and while they are wrong based on the very fundamentals of radiometric dating, that is not the argument we make here.

Simultaneity in this case refers to multiple proposed impact proxies appearing simultaneously in the same sedimentary horizon at the same site, which is not disputed by any of our critics.

Proposed impact proxies are found together in the same layer, and were therefore deposited simultaneously.

Thank you for your diligent work, and sorry for confusing you with all those big words <3

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#8 **Actinopolyspora biskrensis** comment accepted August 2025

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#6

Thank you for responding.

"We will also fix the incorrect chemical formula identified by Elizabeth Bik (sic) above."

Thank you for planning to fix the formula, I had assumed it was merely a typographical error.

FYI - I am not Elisabeth Bik (her first name is spelled with an 's'). I was assigned this username by the PubPeer system when I first created an account. At one time I was anonymous, but in various media stories I have acknowledged my identity as Kevin Patrick, also known on social media as "Cheshire." Research integrity efforts by me are merely a hobby.

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Reply

#9 *Actinopolyspora biskrensis* comment accepted August 2025

#6

Mr. Young - it's astonishing that the journal messed up the references this badly, which must be frustrating for the authors.

Until the journal is able to publish a correction, would you be able to share the correct references here, or post them on a publicly accessible website? I agree that they are "royally messed up," a technical term.

For example, Ref [147] Mahaney W, Krinsley DH, Milner MW, Fischer R, Langworthy K. Did the Black-Mat Impact/Airburst Reach the Antarctic? Evidence from New Mountain Near the Taylor Glacier in the Dry Valley Mountains. J Geol. 2018;126:285-305. <https://doi.org/10.1086/696567> (<https://doi.org/10.1086/696567>)

The correct DOI should be DOI:10.1086/697248

The published DOI 10.1086/696567 resolves to this paper (in the same journal):

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Brahmanism beaten and battered by the Muslim invaders could look to the rulers for support and sustenance and get it. Buddhism beaten and battered by the Muslim invaders had no such hope. It was an uncared for orphan and it withered in the cold blast of the native rulers and was consumed in the fire lit up by the conquerors. (BHIMRAO AMBEDKAR)¹

Scholarship on Buddhist and Muslim interactions has long featured a particular story about Islam's central role in Indian Buddhism's decline. This narrative can be summarized in a single sentence: iconoclastic Muslims invaded India, attacked Buddhist monasteries and institutions of higher learning, and in so doing destroyed the foothold of Buddhism on the subcontinent. It is hard to overstate the potency and prevalence of this story line. As Johan Elverskog has put it, "Whenever the topic of Buddhism and Islam is ever mentioned it almost invariably revolves around the Muslim destruction of the Dharma."² Modern academics often shy away from the more sensational descriptions of this alleged apocalyptic clash that characterized earlier, pop-

For comments and conversations on earlier drafts of this work, I thank Alexandra Kalyanides, Paul Harrison, and the participants of "Buddhist and Muslim Encounters in Premodern South Asia" at the University of Lausanne, October 12-13, 2019.

¹ Quoted in Vasant Moon, ed., *Buddhism in India: Writings and Speeches*, vol. 3 (Bombay: Department of Education, Government of Maharashtra, 1987), 232-33.

² Johan Elverskog, *Buddhism and Islam on the Silk Road* (Philadelphia: University of Pennsylvania Press, 2010), 2.

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This seems similar to the way LLMs sometime hallucinate, although automated reference-linking software could be the culprit too.

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Reply

#10 Marc D. Young comment accepted August 2025

Author Response

#9

Interesting theory regarding AI, I don't have any insight on that other than we certainly don't use it ourselves, but I know it is becoming more common for journals to use in their post-acceptance process.

Here's (https://drive.google.com/file/d/1cpvlpnRAad2y6vSFtgEMXbxxiXZ5wXRr/view?usp=drive_link) a reference list that Chris Moore sent through.

Marc

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Reply

#11 Sinilabeo dero comment accepted August 2025

#7 Thanks for your explanation of your understanding of the difference between synchronicity and simultaneity, but that's not the issue. The problem is that the Bayesian model on which the conclusions are based is flawed. This has been discussed on the PubPeer page about Kennett et al 2015

(<https://pubpeer.com/publications/1F7147A644242D2914CF890FA5F7E0>), which was led by a coauthor of this paper. I notice that the Kennett et al 2015 Bayesian model on which the claims of YDB marker synchronicity (or simultaneity if you prefer that word) was not even cited in this paper. Perhaps that was another mistake by the journal, in which case I look forward to the correction. If it was used, why wasn't it cited? If it wasn't used, why wasn't there discussion of the reason why it has been abandoned?

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Reply

#12 *Actinopolyspora biskrensis* comment accepted August 2025

In the Methods section, this paper says that the radiocarbon ages were calibrated using the ΔR reservoir corrections from Pearce et al. (2023). But Pearce explicitly cautioned that their values “**only remain valid for the pre-bomb, modern situation and most likely large parts of the current Holocene interglacial**,” and that outside the Holocene they “**provide a minimum bound for the calibration of radiocarbon ages**.” Pearce also noted that reservoir ages can vary a lot across the last glacial-interglacial transition because of circulation changes.

Given that, it seems inappropriate to apply Pearce’s modern ΔR values directly to sediments that are ~12,800 years old. If the Bayesian age–depth models in this paper appear more precise than shown by the cited evidence, might the microspherule and platinum peaks actually *not* line up with the onset of the Younger Dryas?

The authors of this paper also used an averaging method that I did not see in Pearce (2023): “Zone 3/2 border for Core 77, $\Delta R = +103$ (avg of +218 and –12).”

Could the authors comment?

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Reply

#13 *Actinopolyspora biskrensis* comment accepted August 2025

In the Methods, the LA-ICP-MS analyses are described as being normalized to Fe = 99%, with other elements calculated relative to that assumption. In the Supporting Information, I only see ratio plots and summary tables that already reflect this normalization.

Could the authors clarify whether the raw, non-normalized LA-ICP-MS data (i.e., the original counts or absolute concentrations before forcing Fe to 99%) are available? Having access to those values would make it possible for others to evaluate how sensitive the conclusions are to the assumption about Fe content.

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Thank you.

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Reply

#15 Marc D. Young comment accepted August 2025

Author Response

#14

Apologies, I thought I had set the permissions properly when I generated the link.

It should work now!

Marc

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Reply

#16 Actinopolyspora biskrensis comment accepted August 2025

#15

Thank you.

Could the authors please make available the original images for Figures 7, 18, and 19, or point me to them?

The images I can find online have inconsistent resolution making them difficult to analyze. I reviewed the supplemental files but did not find them. According to the data availability statement,

"Data availability statement: All data underlying the findings are fully available without restriction. Data are provided within the manuscript and its Supporting Information files, and have been deposited in the Zenodo."

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But perhaps I just haven't located this data. My appreciation in advance.

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#18 *Actinopolyspora biskrensis* comment accepted September 2025

I have two related concerns that I would ask the authors to address.

Regarding Supplemental Table S28: It seems that the choice of modeling all 30 records inside one forced-overlap boundary may be an assumption that directly supports the YDIH. I would like to have a statistics expert determine whether the choice of this model structure itself presupposes the hypothesis. However, after reviewing the information provided in the supplemental files and in the Zenodo repository, it appears that the necessary data to do so has not yet been provided.

Regarding the Data Availability Statement. Currently it reads:

"Data availability statement: *All data underlying the findings are fully available without restriction.* Data are provided within the manuscript and its Supporting Information files, and have been deposited in the Zenodo public repository at: <https://zenodo.org/records/14681287> (<https://zenodo.org/records/14681287>)." (emphasis added)

As noted above, I believe some data necessary to rerun the analysis in Table S28 does *not* seem to be available. Could the authors please provide all the inputs necessary to independently rerun or modify the OxCal modeling? This would include the raw radiocarbon ages (SI only refers to "previously published" sources), the OxCal input script, the calibration details, etc.

Thank you.

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BAYESIAN SYNCHRONITY TEST

Table S28. Calculations for testing synchronicity (16, 124). Dates from 30 sites: 1 site with the Greenland platinum record, 6 for Younger Dryas climate onset, and 23 for the YDB (grouped by quality). The Bayesian modeled ages for the 30 records are separated by a minimum of zero (statistically calculated as "-5") years at 68%, 95%, and 99% ("Difference L" row in green). The maximum statistical difference is 130 years at 95% CI. Using the Date code, the length of the overlap among 30 sites was modeled from the Start and End Boundaries (two green highlighted dates), between which all dates fall, producing a modeled age range of 12,820 to 12,740 Cal B.P. at 95% (bottom row highlighted in green). Age model is in Fig. 9 in the main manuscript.

Difference Code: modeled YDB ages and layers	Prior ages μ σ	Common μ σ	95.4% range	68.2% range	95.4% range	99.7% range	A model=188 A overall=190.7
Difference L		45 40		-5 60	-5 130	-5 215	
Boundary End		12760 25		12790 12740	12805 12710	12820 12665	
AGE OF YOUNGER DRYAS ONSET							
C_Date GRIP GICC05 ice model	12845 140	12785 25	13125 12570	12810 12760	12840 12735	12885 12700	127
C_Date GISP2 ice model	12840 260	12785 25	13360 12320	12810 12760	12835 12735	12885 12695	137.7
C_Date GISP2 platinum peak	12840 260	12785 25	13360 12315	12810 12760	12835 12735	12885 12695	138.1
C_Date Cariaco Basin varves	12770 30	12780 20	12835 12710	12805 12760	12820 12740	12840 12720	114.9
R_Date Late glacial tree-rings, GER	12810 50	12785 20	12920 12720	12810 12765	12830 12745	12865 12725	126.5
C_Date Hulu speleothems, CHN	12770 60	12785 20	12895 12650	12805 12760	12830 12735	12865 12705	130.5
C_Date Meerfelder Maar	12680 125	12785 25	12935 12425	12805 12760	12835 12735	12875 12695	101.4
YDB: HIGH QUALITY							
Prior Abu_Hureyra_UCIAMS_105429	12825 55	12790 25	12935 12705	12810 12765	12840 12740	12880 12715	92.1
Prior Arlington_Cyn_YDB_layer	12805 55	12785 20	12925 12695	12810 12765	12835 12740	12875 12715	129.1
Prior Aalsterhut_YDB_layer	12780 35	12780 20	12845 12725	12800 12760	12820 12745	12845 12730	106.5
Prior Big_Eddy_YDB_layer	12770 85	12785 20	12935 12580	12805 12760	12835 12740	12870 12705	143.4
Prior Bull_Creek_YDB_layer	12840 75	12790 20	12995 12710	12810 12765	12835 12740	12880 12715	122.3
Prior Daisy_Cave_YDB_age	12730 320	12785 25	13320 12050	12810 12760	12840 12735	12885 12695	141.4
Prior Lake_Hind_YDB_UCIAMS_29317	12745 180	12785 25	13190 12550	12810 12760	12835 12730	12880 12685	37.8
Prior Lingen_YDB_Beta_369246	12735 85	12780 20	12910 12520	12805 12760	12820 12730	12855 12700	87.7
Prior Sheriden_Cave_YDB_layer	12840 120	12785 20	13110 12625	12805 12765	12835 12740	12880 12710	146.1
YDB: MEDIUM QUALITY							
Prior Barber_Creek_YDB_age_UW_1908	12865 535	12785 25	13945 11865	12810 12760	12835 12735	12885 12695	138.4
Prior Blackwater_YDB_age	12775 365	12785 25	13510 12090	12810 12760	12840 12735	12885 12695	135
Prior Indian_Creek_YDB_age	12750 425	12785 25	13495 11805	12810 12760	12840 12735	12890 12700	120.8
Prior Lindenmeier_YDB_L_141	12775 180	12785 25	13195 12440	12805 12760	12835 12735	12880 12695	131.4
Prior Murray_Springs_YDB_layer	12750 235	12785 25	13195 12255	12810 12760	12840 12735	12885 12700	147.4
Prior Santa_Maira_YDB_Beta-75225	12785 295	12785 25	13265 12070	12805 12760	12835 12735	12885 12705	147.6
Prior Talega_YDB_Beta_196150	12860 150	12790 25	13075 12545	12810 12765	12845 12740	12890 12710	62.7
Prior Topper_YDB_AA100294	12785 185	12785 20	13085 12365	12805 12760	12835 12735	12880 12710	158.9
YDB: LOWER QUALITY							
Prior Blackville_YDB_age_LB859	12820 1080	12785 25	15015 10705	12805 12760	12835 12735	12885 12695	142.8
Prior Lake_Cuitzeo_YDB_layer	12850 570	12785 25	14265 12195	12805 12760	12835 12730	12880 12690	60.3
Prior Lommel_YDB_age	12735 790	12785 25	14410 11325	12810 12760	12835 12735	12885 12695	120.4
Prior Melrose_YDB_LB860b	12255 2405	12785 25	17185 7710	12810 12760	12840 12735	12885 12695	129.2
Prior Mucunuque_YDB_age	12845 630	12785 25	13550 11335	12810 12760	12840 12735	12890 12700	59.7
Prior Ommen_YDB_age	12750 560	12785 25	13605 11425	12810 12760	12840 12735	12885 12695	94.3
Phase							
Boundary Start		12810 25		12830 12775	12865 12760	12920 12745	
Sequence YDB SYNCHRONICITY							
Prior Boundary Start to Boundary End							
Synchronicity_layer		12785 25		12805 12760	12835 12735	12880 12700	

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Reply

#19 Mark Boslough comment accepted September 2025

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In the current online version of Moore et al., creation [100] is this:

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Boslough M. In: 2015.

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Citation [169] in the "corrected" version is this:

Boslough M, editor Airburst Modeling. First International Workshop on Potentially Hazardous Asteroids Characterization, Atmospheric Entry and Risk Assessment,; 2015; NASA Ames Research Center, Mountain View, CA: Sandia National Lab.(SNLNM), Albuquerque, NM (United States);

I was not an editor of this workshop and there were no proceedings published as far as I know. I also see a punctuation error. This mistake seems to have crept into Google Scholar from previous incorrect citations to it by other papers by the Comet Research Group, only to be found by the chatbot and propagated into the current paper.

I think ChatGPT was struggling but failed to find the paper you surely intended to cite, which was the basis for the slides. The main point of my presentation was to summarize a book chapter called "Airburst Modeling" (published the same year) and to show animated versions of some of the simulations. Here's the correct citation:

Boslough M. Airburst Modeling, 2015. In Handbook of cosmic hazards and planetary defense, J.N. Pelton, F. Allahdadi (eds) pp. 665-692. Springer. DOI 10.1007/978-3-319-03952-7_56

I would appreciate it if you make this correction in the present paper as well as future papers by the CRG.

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Reply

#20 **Mark Boslough** comment accepted September 2025

My previous comment (**#19**) may now appear to be a non-sequitur that requires some explanation. It was my reply to Marc Young's comment (**#17**), which has now been deleted. As it turns out, this paper made use of ChatGPT "to assist with language refinement and clarity improvements during manuscript preparation". It appears that some of the citations, such as [169] to a nonexistent publication by me, may therefore be attributable to AI hallucinations. I hope the authors and editors can get this problem cleared up. Here's a screenshot of the paper's acknowledgment that ChatGPT was used in the preparation of this paper, including (apparently) the citations.

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Acknowledgments

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#21 Mark Boslough comment accepted December 2025

Two coauthors of this paper (West and Kennett) were also coauthors of Kurbatov (2010), which is citation [84]. This paper states,

Since 2007, numerous publications have presented evidence consistent with the YDIH from sites on six continents. This evidence includes anomalously high concentrations of multiple impact-related proxies, including anomalies in platinum [8,9,50,78–83], iridium [1,9,80,84,85], magnetic & non-magnetic microspherules [1,8,9,42,43,81–83,86,87], meltglass [8,9,47,73], nanodiamonds [8–10,80,88–90], and combustion aerosols and soot [8–10,49,75,76,79,80]. These proxies have been reported in a sediment layer dating to the lower Younger Dryas Boundary (YDB) across North America [1], central Mexico [91], South America [92,93], South Africa [94], Europe [1,87,95–97], Greenland [84], the Middle East [8–10,80] and in lacustrine and ice core records globally [75,76].

It would appear that the authors are backing away from their past claims that Kurbatov et al 2010 discovered evidence consistent with the YDIH in the form anomalously high concentrations of nanodiamonds, because a

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#22 Mark Boslough comment accepted December 2025

I noticed a couple other mistakes in the paragraph quoted in my previous comment (#21). The authors also cited Kurbatov et (2010) as presenting evidence consistent with anomalously high concentrations of iridium, but that is not true. Kurbatov et al only mention iridium one time in the entire paper. It is in their "future work" subsection. One item in a list of future areas of potential exploration is "more detailed geochemical time-series measurements, including for osmium and iridium".

The other mistake is the claim that publications have presented evidence from sites on six continents. It lists five continents either by name or by reference to places within them: North America, South America, Africa, Europe, and Asia. It also lists Greenland, but Greenland is not a continent.

I hope the authors will consider making corrections to these factual errors.

#23 Hoya camphorifolia comment accepted February 2026

Retracted 11 February 2026 (<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0342613>).

Following publication of this article [1], concerns were raised regarding referencing, methodology, and data reporting in this article. The PLOS One Editors have consulted multiple experts regarding these concerns. The following issues with this article have been identified:

- Incorrect citations following an issue with the DOI information provided by the authors, raising concerns about undisclosed use of AI tools.
- Multiple citations are incorrectly used and do not support the associated statements made.
- Several concerns have been raised about the reported chronology. These relate to (a) issues with reporting of the methods used to generate new dates; and (b) issues with calibration leading to potentially erroneously older ages. Concerns raised include, but are not limited to:

- Missing information regarding the foraminifera species or type used for dating, and the number of individuals used.

Lack of clarity on how authors accounted for the uncertainty of producing dates from 5 cm sediment (https://chrome.google.com/webstore/detail/pubpeer/fmcdfigcdfkdghdkblbbpacikscbbbh) | Blog (https://pubpeer.com/blog) | Journals (https://journals.plos.org/pubpeer/) | Institutions (https://pubpeer.com/institutions) | About (https://pubpeer.com/static/about) | Privacy Policy (https://pubpeer.com/static/privacy)

- The method for calibrating the radiocarbon ages does not follow the recommended methods for the latitude of the study site and time period. Calibrating radiocarbon ages using the recommended methods produces ages that are younger than those reported in the article and that are outside the Younger Dryas.
- It is unclear if presented ages are calibrated or modelled.
- Misidentification of material in SI Figs S30–S34 (<https://zenodo.org/records/14681287>) (<https://zenodo.org/records/14681287>). The authors claim these are spherule fragments identified as cosmic impact indicators. During expert consultation, these have been identified as marine foraminifera. The presence of foraminifera will affect the Ca curves presented and raise concerns with the peak abundance of cosmic material identified in each core, as some of this may be from foraminifera presence. S32 and S34 have clearly visible micropores and a chemical composition of Ca, C, O. Furthermore, it is unknown how much of the reported ‘Cosmic indicator spike’ includes this misidentified material.
- Sampling methodology does not allow for long term variability in cosmic particle concentration to be assessed. The peak at core 64 of a single data point is outside the presented Younger Dryas Boundary, but of the same magnitude as the peak at the Younger Dryas Boundary in core 67.
- The article does not position itself within wider literature, with no discussion as to why the mainstream hypothesis is refuted.
- The data in the Zenodo data repository is not always clearly attributed.

In light of the above concerns that call into question the validity of the reported results and the conclusions made, the PLOS One Editors retract this article. PLOS regrets that the issues were not identified prior to the article’s publication.

JJ agreed with the retraction. CM, JK, AA, KL, SC, GK, AW, and ML did not agree with the retraction. VAT responded but expressed neither agreement nor disagreement with the editorial decision. DM, MB, WN, MB, VA, SS, JP, TW, MY, MA, RG, and JM either did not respond directly or could not be reached.

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#24 Mark Boslough comment accepted February 2026

There is a large set of PubPeer pages about papers on the same topic (the Younger Dryas impact hypothesis, or YDIH). Many of them have similar or related issues and the PubPeer pages provide further context. They can be accessed here (<https://pubpeer.com/search?q=ydih>).

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#25 Couratari guianensis comment accepted March 2026

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See also <https://retractionwatch.com/2026/03/04/controversial-comet-theory-struck-by-two-new-retractions/>
(<https://retractionwatch.com/2026/03/04/controversial-comet-theory-struck-by-two-new-retractions/>)

<https://pubpeer.com/publications/6E3912F0B572851E0CD76504A8351A>
(<https://pubpeer.com/publications/6E3912F0B572851E0CD76504A8351A>) (the other paper).

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